

CATALYSIS

Selective conversion of syngas to light olefins

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Although considerable progress has been made in direct synthesis gas (syngas) conversion to light olefins ($C_2=C_4$) via Fischer-Tropsch synthesis (FTS), the wide product distribution remains a challenge, with a theoretical limit of only 58% for C_2-C_4 hydrocarbons. We present a process that reaches $C_2=C_4$ selectivity as high as 80% and C_2-C_4 94% at carbon monoxide (CO) conversion of 17%. This is enabled by a bifunctional catalyst affording two types of active sites with complementary properties. The partially reduced oxide surface ($ZnCrO_x$) activates CO and H_2 , and C-C coupling is subsequently manipulated within the confined acidic pores of zeolites. No obvious deactivation is observed within 110 hours. Furthermore, this composite catalyst and the process may allow use of coal- and biomass-derived syngas with a low H_2/CO ratio.

Fischer-Tropsch synthesis (FTS) has played an important role as a gas-to-liquid technology, producing synthetic lubricants and synthetic fuels from coal, natural gas, or biomass since it was developed almost a century ago. It is also the only effective technology to date for direct conversion of synthesis gas (syngas) to light olefins—i.e., $C_2=C_4$ —olefins containing two to four carbon atoms, so called Fischer-Tropsch to olefins (FTTO) (1–5). FTS has been under extensive study for more than 50 years, and a variety of metal catalysts, including iron, cobalt,

and ruthenium, have been tested (6). However, application of this technology is still limited by low olefin selectivity and high methane selectivity, as well as severe carbon deposition. These drawbacks arise from the FTS reaction mechanism, which is generally accepted to proceed via surface polymerization of CH_x ($x = 1, 2$, or 3)—i.e., addition of CH_x monomer units to the adsorbed alkyl species on open metal surfaces. Thus, a wide distribution of hydrocarbons with different chain lengths is produced. It can be described by the Anderson-Schulz-Flory (ASF) model (7, 8), which predicts that selectivity of C_2 to C_4 hydrocarbons (C_2-C_4), including $C_2=C_4$ olefins and C_2-C_4 paraffins, does not exceed 58% (6).

Thus, the key challenge of selective formation of light olefins from syngas lies in precise control of C-C coupling while suppressing overhydrogenation and methane formation. We present here a process named OX-ZEO (Oxide-Zeolite),

which separates CO activation and C-C coupling onto two different types of active sites with complementary properties. CO and H_2 are activated over a partially reduced oxide ($ZnCrO_x$) surface, whereas C-C coupling is controlled within the confined environment of zeolite pores with acidic sites. As a result, a C_2-C_4 selectivity up to 94% (including 80% $C_2=C_4$ and 14% C_2-C_4) of all hydrocarbons (carbon atom-based) was achieved with only 2% methane at a CO conversion of 17%. This C_2-C_4 selectivity is far beyond the maximum predicted by the ASF model in FTS.

The composite catalyst contained an oxide ($ZnCrO_x$) that exhibits a typical spinel structure (fig. S1) and a mesoporous SAPO zeolite (MSAPO) exhibiting CHA structure with a hierarchical pore texture (figs. S1 and S2 and table S1) (9). Figure 1A shows that $C_2=C_4$ selectivity reached 74% at CO conversion of 16% over a catalyst with a mass ratio of $ZnCrO_x/MSAPO = 1.4$, under reaction conditions of $H_2/CO = 1.5$, pressure 2.5 MPa, and space velocity 4800 ml/h·g_{cat}. A higher H_2/CO ratio benefits CO conversion, which rose to 30% at $H_2/CO = 3$, for instance, whereas a higher space velocity favored the selective formation of olefins. Selectivities varied in the range from 67 to 80% for $C_2=C_4$ and 81 to 94% for C_2-C_4 in the studied H_2/CO (0.5 to 3.0) and space velocity ranges (1285 to 7714 ml/h·g_{cat}) and different ratios of $ZnCrO_x/MSAPO$. This $C_2=C_4$ selectivity is higher than the best value reported for FTTO (61%) (4) and exceeds the maximum predicted for C_2-C_4 hydrocarbons according to the ASF distribution in typical FTS, as depicted in Fig. 1B. Both CH_4 and C_5+ selectivities were <5% (Fig. 1A), prominently lower than those in FTTO (in a range of 10 to 40% for CH_4 and >10% for C_5+) (4). In contrast, the C_1 and C_5+ products were difficult to suppress simultaneously in FTTO; causing one to decline caused the other to increase (1, 7). The reactivity of this composite catalyst showed good reproducibility, with CO conversion and selectivity fluctuating within 4 and 5%, respectively, among 11 tests (fig. S3). Furthermore, it delivered rather

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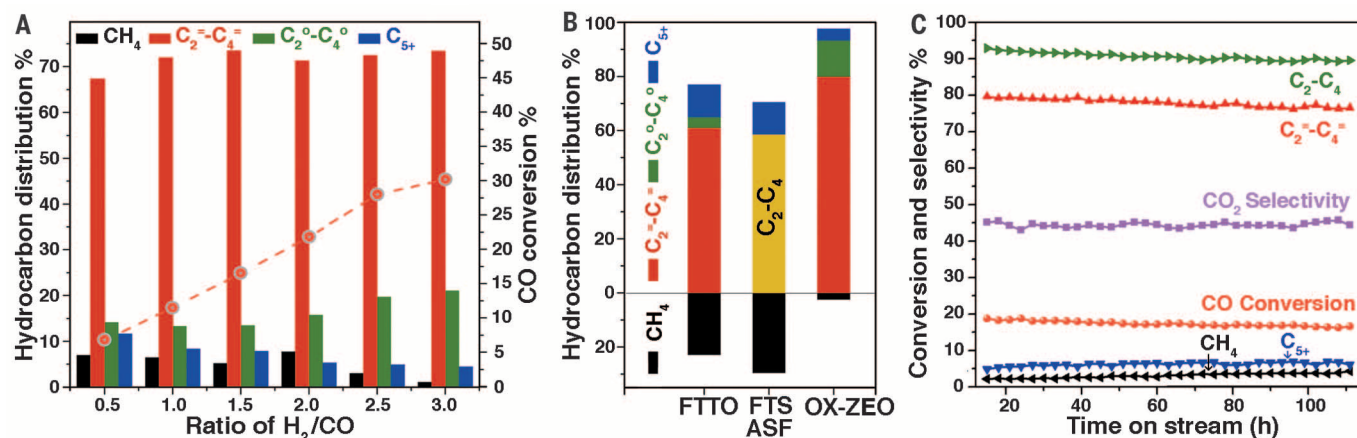


Fig. 1. Catalytic process of OX-ZEO. (A) CO conversion and product distribution at different H_2/CO ratios in syngas over a catalyst with a mass ratio of $ZnCrO_x/MSAPO = 1.4$ at a space velocity of 4800 ml/h·g_{cat}. (B) Hydrocarbon distribution in OX-ZEO in comparison to that reported for FTTO (4) and that in FTS predicted by the ASF model at a chain growth probability of 0.46, with the yellow bar representing selectivity of C_2-C_4 hydrocarbons. (C) A stability test of a composite catalyst with $ZnCrO_x/MSAPO$ ratio = 0.9 at 6828 ml/h·g_{cat} and H_2/CO of 2.5.

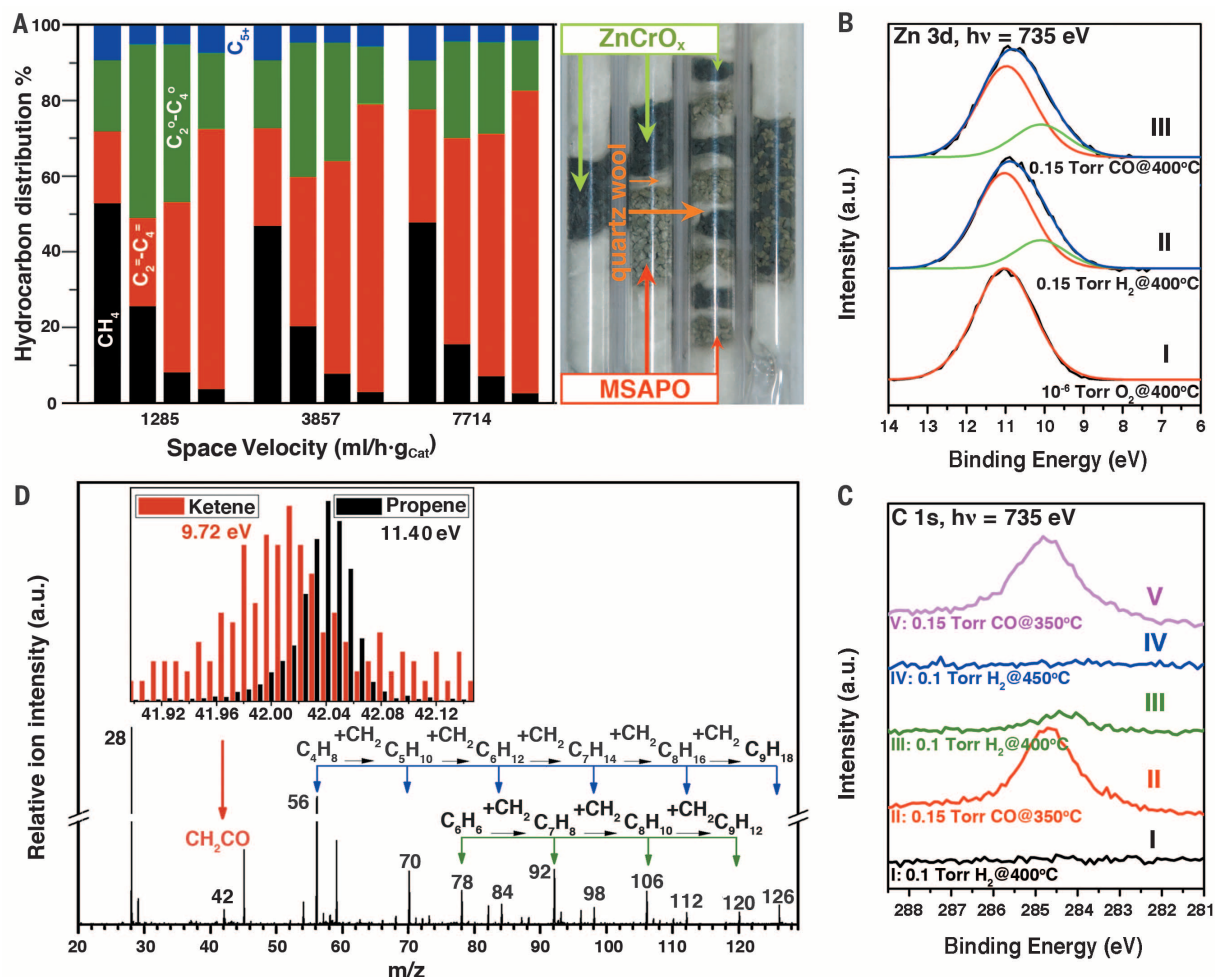


Fig. 2. Bifunctionality of the composite ZnCrO_x/MSAPO catalyst and investigation of the reaction intermediate. (A) Reaction results over the catalysts with the two functionalities packed in different modes under the same conditions. The picture on the right side displays the catalyst beds after reaction, representing Mode 1 to 4 from left to right: Mode 1, the catalyst contains ZnCrO_x only; Mode 2, MSAPO packed below ZnCrO_x, separated by an inert layer of quartz wool; Mode 3, MSAPO and ZnCrO_x packed in an alternating

sequence and separated by quartz wool; Mode 4, MSAPO and ZnCrO_x well mixed. (B) In situ NAP-XPS Zn3d spectra of ZnCrO_x under different conditions. (C) In situ NAP-XPS C1s spectra of ZnCrO_x exposed to H₂, CO, and H₂ sequentially, and then again to CO atmosphere under different conditions. (D) In situ study of syngas conversion over ZnCrO_x by SVUV-PIMS at hν = 9.72 eV. The insets display the signals of m/z = 42.01 (ketene) and m/z = 42.05 (propene) detected at hν = 9.72 and 11.40 eV, respectively.

stable performance (Fig. 1C). The total C₂–C₄ selectivity remained >90%, and that of C₂[–]–C₄[–] ~78% during a 110-hour test at 400°C, 2.5 MPa, and a space velocity of 6828 ml/h·g_{cat}.

Although composite catalysts containing metal oxides such as Cr₂O₃–ZnO and CuO–ZnO, and zeolites such as ZSM-5, Y, and β have been attempted previously for syngas conversion (10–15), the products were mainly dimethylether (10), liquefied petroleum gas (C₃^o–C₄^o) (11–13), or gasoline (14, 15). We attribute the efficiency of the OX-ZEO process to the bifunctionality of the catalyst, with two types of active sites exhibiting complementary and compatible properties. In the absence of MSAPO (corresponding to Mode 1 of Fig. 2A), syngas was converted mainly to CH₄ (selectivity 53%), and selectivity to C₂–C₄ hydrocarbons was only 38% over ZnCrO_x. Upon combination with MSAPO, which was packed below the oxide and separated by a layer of inert quartz wool (Mode 2), the products shifted and C₂–C₄ selectivity increased to

69% containing 23% C₂[–]–C₄[–], whereas CH₄ selectivity dropped to 26%.

This result shows that reaction intermediates generated over the oxides must have transported in the gas phase toward the active sites of MSAPO, where they were converted to C₂–C₄ hydrocarbons, rather than being hydrogenated to CH₄ over the oxide or in gas phase. Thus, any measures that could facilitate the transport of intermediates in gas phase should benefit selective formation of olefins, for example, increasing space velocities, as demonstrated in Fig. 2A. In addition, Mode 3, where the oxides and MSAPO were packed in an alternating sequence and thus the transport distance was reduced, also leads to enhanced C₂[–]–C₄[–] selectivity (65% at 7714 ml/h·g_{cat}). We observed an even higher selectivity of C₂[–]–C₄[–] (80%), with overall C₂–C₄ selectivity reaching 94% and only 2% CH₄ over the well-mixed composite catalyst (Mode 4). Furthermore, CO conversion could be tuned by varying the relative mass ratio of oxide/

zeolite (fig. S4) (9), which lends further support to the above hypothesis that the composite catalyst is bifunctional and that the reaction involves intermediate transport in gas phase.

Activation of the C–O bond has been reported previously for surfaces promoted with oxygen vacancies—for example, Fe₃O₄ (16), CoO_x/TiO₂ (17), Cu/ZnO (0001) (18), and ZnGa₂O₄ (19). However, little is known about subsequent formation of C–C and C–H bonds in the presence of H₂. Density functional theory (DFT) calculations using the ZnCr₂O₄ spinel (111) surface as a model reveal that it is reducible, resulting in a surface with a number of oxygen vacancies (figs. S5 and S6) (9). In situ near-ambient pressure x-ray photoelectron spectroscopy (NAP-XPS) of ZnCrO_x surface (Fig. 2B) revealed a new signal with a lower Zn3d binding energy upon exposure to H₂ and CO, compared with the initial oxidized surface. CO can be activated on such a surface, leading to formation of CO₂, detected by time-of-flight mass spectrometry

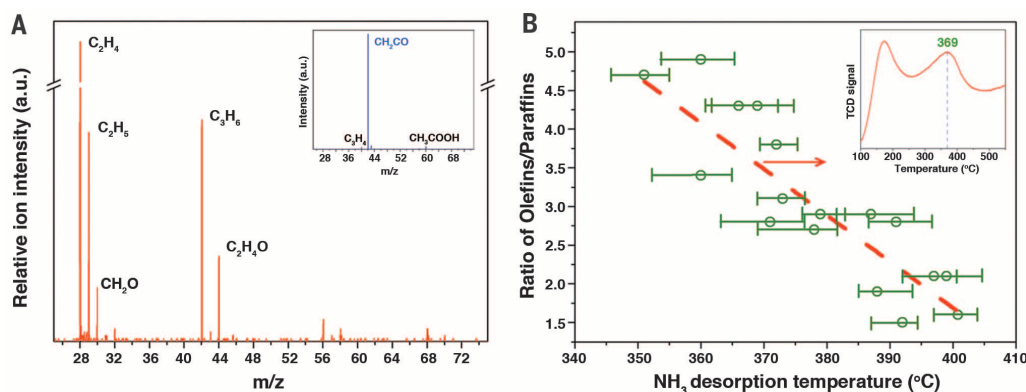


Fig. 3. The role of MSAPO in OX-ZEO.

(A) Catalytic conversion of ketene to olefins over MSAPO at 400°C, with the effluent monitored by SVUV-PIMS. The inset shows the results of the blank experiment with ketene fed to the same reactor containing no catalyst. Spectra are recorded at 10.70 eV. (B) Ratio of olefin/paraffin as a function of the NH_3 desorption peak temperature from the medium-strength acidic sites of MSAPO. The dashed line is only for the purpose of guiding the eyes, and the inset displays a typical NH_3 -TPD profile.

(TOF-MS) (fig. S7) (9) and surface $^*\text{C}$ species in absence of hydrogen, which is evidenced by a strong C1s signal emerging at 284.7 eV, even at 350°C (curve II in Fig. 2C). This C1s signal remained even after heat treatment at 300°C under ultrahigh vacuum (fig. S8) (9), but it remarkably attenuated in H_2 (curve III) and finally vanished at 450°C (curve IV). Analysis of the effluent by TOF-MS shows that $^*\text{C}$ species were removed via hydrogenation forming hydrocarbons (fig. S7, b to d) (9). Thus, the precursor of hydrocarbons, CH_x species formed over ZnCrO_x , are likely the intermediates in the OX-ZEO process.

To detect the intermediate in gas phase, we turned to the highly sensitive synchrotron-based vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) (9), which has been widely used in the field of combustion to detect active intermediates and radicals with low concentrations in gas phase (20, 21). It provides a tunable photon energy, soft ionization, and superior signal-to-noise ratio. The schemes of SVUV-PIMS and the reactor are shown in fig. S9. When syngas was fed into the reactor in the absence of catalysts (a blank experiment), no other signal was observed. In the presence of ZnCrO_x , a signal of mass/charge ratio (m/z) = 42.01 appeared in the effluents at a photon energy $h\nu$ = 9.72 eV (Fig. 2D) in addition to stable hydrocarbon products. Further varying the photon energy (fig. S10) (9) confirmed that this signal was unambiguously attributed to ketene CH_2CO , considering the m/z ratio and the ionization energy (22–24). In comparison, propene (m/z = 42.05) was only distinguishable at $h\nu$ > 9.73 eV (fig. S10e) (9), and it became prominent at $h\nu$ = 11.40 eV (the black signal in the inset of Fig. 2D). Additional signals in Fig. 2D were assigned to stable products of C_4H_8 and C_6H_6 , and their derivatives from stepwise addition of CH_2 monomers up to m/z = 126.

In addition to ketene and the stable products, signals of methanol (m/z = 32) and its dissociated product methoxyl group (m/z = 31) were also detected at a high $h\nu$ (11.40 eV) in the presence of ZnCrO_x catalyst. However, the reaction did not appear to go via methanol in this OX-ZEO process because the yield of C_2^- – C_4^- dramatically dropped to 3% within 22 hours when feeding methanol (50 mbar in He) directly to the composite catalyst (fig. S11). Figure S12 shows that

MSAPO alone as a methanol-to-olefin (MTO) catalyst also deactivated quickly within 30 hours, even at a methanol partial pressure as low as 5 mbar, and 0.18 mol C/g_{cat} was converted before deactivation. By contrast, the composite catalyst delivered rather stable performance in OX-ZEO. No obvious degradation of CO conversion was observed for operation as long as 650 hours (fig. S13), and 9.6 mol C/g_{cat} was converted within this period of time. The total coke deposit was only 11 weight % (wt %) (fig. S14) (9). In contrast, fast deactivation remains a major issue for MTO in that coke deposits can mount up about 10 wt % within 15 min (25). Furthermore, very little water is produced in OX-ZEO, in contrast to the MTO process, where two moles of H_2O are produced for each mole of ethylene (assumed as the only product) from methanol (9). In addition, O from CO is removed mainly by producing CO_2 with a selectivity of 40 to 45% in OX-ZEO, whereas little CO_2 forms in MTO. The above results show that reaction via methanol may not be the dominating pathway in OX-ZEO, although it cannot be excluded completely, whereas ketene likely plays an important role.

Such CH_2 species are very active and readily react with CO in the presence of CO, forming a relatively less reactive ketene (26, 27), which is detectable by SVUV-PIMS. Thus, CH_2 species are likely the primary intermediates. By forming ketene, the reaction pathway of surface polymerization of CH_x is blocked, hence circumventing the ASF limits of FTS. Subsequently, in the presence of confined acidic environment of zeolite pores, CH_2CO can be converted to olefins (Fig. 2A). This was validated by feeding ketene directly as the reactant to the MSAPO catalyst, with the effluents monitored by the online SVUV-PIMS. Ketene was synthesized via pyrolysis of acetic anhydride following the reported procedure (fig. S15) (24, 28). Comparison of the results in the blank reactor and in the presence of the catalyst (Fig. 3A and fig. S15) demonstrates the capability of MSAPO catalyzing conversion of ketene to light olefins.

Figure 3B and fig. S16 (9) demonstrate that the product selectivity can be modulated by the medium strength acidity, characterized by temperature-programmed-desorption (TPD) of NH_3 , with the peak maximum located in the range from 350° to 410°C. For example, a commercially available

SAPO-34 exhibiting NH_3 desorption at ~394°C had a C_2^- – C_4^- selectivity of 43% and an olefin/paraffin ratio (C_2^- – $\text{C}_4^-/\text{C}_5^-$ – C_6^-) of 0.9. The olefin/paraffin ratio increased with the decreasing NH_3 desorption temperature and reached 4.7 for a MSAPO sample with NH_3 desorption peak at 350°C (Fig. 3C). In contrast, a composite catalyst with almost no medium strength acidic sites had a similar product distribution, as the catalyst contained no zeolite (fig. S17 and table S3) (9). Thus, it is reasonable to assume that there may exist an optimum acidity with NH_3 desorption temperature <350°C for olefin formation. However, it is still a challenge to synthesize zeolites with a weaker acidity but without reducing the number of active sites (necessary to achieve a reasonable conversion). In addition, the preliminary results in fig. S18 (9) show that the distribution of different-sized hydrocarbons might be tunable through the shape selectivity of zeolites, because a larger pore generally yields higher hydrocarbons. However, further elucidation of the relation between the structure (pore size/acidity) and the activity/selectivity will require much more sophisticated experiments because the acidity frequently varies simultaneously with the pore structure and crystallinity.

The capability of the partially reduced oxide in activating CO but incapability of catalyzing surface polymerization of CH_x makes it possible to manipulate C–C coupling with confined acidic zeolite pores. Thus, the CO conversion and selectivity can be tuned at the same time—i.e., CO conversion is manipulated via the surface structure of the oxides and the ratio of oxides/zeolite, whereas the olefin selectivity is controlled by the properties of zeolites, particularly the pore structure and acidity. These findings open up a new avenue for development of syngas-to-olefin technologies, which may allow utilization of coal- and biomass-derived syngas with a low H_2/CO ratio.

REFERENCES AND NOTES

1. R. Snel, *Catal. Rev. Sci. Eng.* **29**, 361–445 (1987).
2. Y. D. Xu, D. W. Goodman, *Catal. Lett.* **24**, 31–35 (1994).
3. Z. Yang, S. Guo, X. Pan, J. Wang, X. Bao, *Energy Environ. Sci.* **4**, 4500–4503 (2011).
4. H. M. Torres Galvis *et al.*, *Science* **335**, 835–838 (2012).
5. X. Chen, D. Deng, X. Pan, Y. Hu, X. Bao, *Chem. Commun. (Camb.)* **51**, 217–220 (2015).
6. H. M. Torres Galvis, K. P. de Jong, *ACS Catal.* **3**, 2130–2149 (2013).

7. R. A. Friedel, R. B. Anderson, *J. Am. Chem. Soc.* **72**, 1212–1215 (1950).
8. I. Puskas, R. S. Hurlbut, *Catal. Today* **84**, 99–109 (2003).
9. Supplementary materials are available on Science Online.
10. G. Yang, N. Tsubaki, J. Shamoto, Y. Yoneyama, Y. Zhang, *J. Am. Chem. Soc.* **132**, 8129–8136 (2010).
11. K. Fujimoto, H. Saima, H. O. Tominaga, *J. Catal.* **94**, 16–23 (1985).
12. Q. Ge, X. Li, H. Kaneko, K. Fujimoto, *J. Mol. Catal. Chem.* **278**, 215–219 (2007).
13. Y. Yu et al., *React. Kinet. Mech. Catal.* **112**, 489–497 (2014).
14. J. Erenia, J. M. Arandes, J. Bilbao, A. T. Aguayo, H. I. de Lasa, *Ind. Eng. Chem. Res.* **37**, 1211–1219 (1998).
15. P. Mohanty, K. K. Pant, J. Parikh, D. K. Sharma, *Fuel Process. Technol.* **92**, 600–608 (2011).
16. Y. Tamaura, M. Tahata, *Nature* **346**, 255–256 (1990).
17. G. Melae et al., *J. Am. Chem. Soc.* **136**, 2260–2263 (2014).
18. J. Xiao, T. Frauenheim, *J. Phys. Chem. Lett.* **3**, 2638–2642 (2012).
19. C. Jia et al., *Phys. Chem. Chem. Phys.* **16**, 7538–7547 (2014).
20. F. Qi, *Proc. Combust. Inst.* **34**, 33–63 (2013).
21. J. D. Savee et al., *Science* **347**, 643–646 (2015).
22. NIST Chemistry WebBook, <http://webbook.nist.gov/chemistry/mw-ser.html>.
23. T. A. Cool et al., *J. Chem. Phys.* **119**, 8356–8365 (2003).
24. B. Yang et al., *Int. J. Mass Spectrom.* **309**, 118–128 (2012).
25. D. Chen, K. Moljord, A. Holmen, *Micro. Meso. Mater.* **164**, 239–250 (2012).
26. J. Channugam, M. Burton, *J. Am. Chem. Soc.* **78**, 509–519 (1956).
27. Z. H. Zhang, Y. Zhang, J. B. Wang, *ACS Catal.* **1**, 1621–1630 (2011).
28. J. W. Williams, C. D. Hurd, *J. Org. Chem.* **5**, 122–125 (1940).

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SUPPLEMENTARY MATERIALS

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Materials and Methods
Figs. S1 to S18
Tables S1 to S4
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QUANTUM COMPUTING

Realization of a scalable Shor algorithm

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Certain algorithms for quantum computers are able to outperform their classical counterparts. In 1994, Peter Shor came up with a quantum algorithm that calculates the prime factors of a large number vastly more efficiently than a classical computer. For general scalability of such algorithms, hardware, quantum error correction, and the algorithmic realization itself need to be extensible. Here we present the realization of a scalable Shor algorithm, as proposed by Kitaev. We factor the number 15 by effectively employing and controlling seven qubits and four “cache qubits” and by implementing generalized arithmetic operations, known as modular multipliers. This algorithm has been realized scalably within an ion-trap quantum computer and returns the correct factors with a confidence level exceeding 99%.

Shor’s algorithm for factoring integers (*1*) is one example in which a quantum computer (QC) outperforms the most efficient known classical algorithms. Experimentally, its implementation is highly demanding (*2–7*) because it requires both a sufficiently large quantum register and high-fidelity control. Such challenging requirements raise the question of whether optimizations and experimental shortcuts are possible. Optimizations, especially system-specific or architectural optimizations, are certainly possible, but for a demonstration of Shor’s algorithm in a scalable manner, special care must be taken to not oversimplify the implementation—for instance, by employing knowledge about the solution before the actual experimental application (*8*).

How does Shor’s algorithm work? First, we consider a classical factoring recipe, assuming

that the number we want to factor is $N = 15$. We pick a random number $a \in [2, N - 1]$ (the base)—say, $a = 7$. We evaluate whether the greatest common divisor $\text{gcd}(a, N) = 1$; if not, a factor is already determined. This is the case for $a = \{3, 5, 6, 9, 10, 12\}$. Next, we calculate the modular exponentiation $a^x \bmod N$ for $x = 0, 1, 2, \dots$ and find its period r : the first value of $x > 0$ such that $a^x \bmod N = 1$. Given r , finding the factors of N requires calculating the greatest common divisors of $a^{r/2} \pm 1$ and N , which is efficiently possible with a classical approach—for instance, using Euclid’s algorithm. For our example ($N = 15$, $a = 7$), the modular exponentiation yields 1, 7, 4, 13, 1, ..., which has a period of 4. The greatest common divisors of $a^{r/2} \pm 1 = 7^{4/2} \pm 1 = \{48, 50\}$ and $N = 15$ are $\{3, 5\}$, the nontrivial factors of N . In this example, the cases $a = \{4, 11, 14\}$ have period $r = 2$ and require a single multiplication step ($a^2 \bmod N = 1$), which is considered an “easy” case (*8*). Note that the periodicity for a chosen a cannot be predicted.

How can this recipe be implemented in a QC? A QC also has to calculate $a^x \bmod N$ in a computational register for $x = 0, 1, 2, \dots$ and then extract r . Using the quantum Fourier transform (QFT) applied to the period register, the period of

$a^x \bmod N$ can be extracted from a number of measurements not increasing with the size of the number to be factored.

What are the requirements and challenges of implementing Shor’s algorithm? We first focus on the period register and subsequently address modular exponentiation in the computational register. Factoring N , an $n = \lceil \log_2(N) \rceil$ -bit number (with the quantity in brackets rounded up to next integer number), requires a minimum of n qubits in the computational register (to store the results of $a^x \bmod N$) and generally about $2n$ qubits in the period register (*9, 10*). Thus, even a seemingly simple example, such as factoring 15 (an $n = 4$ -bit number), requires $3n = 12$ qubits. These qubits then have to be manipulated with high-fidelity gate operations. Given the current state-of-the-art control over quantum systems (*11*), such an approach would probably yield an unsatisfactory performance. However, a full quantum implementation of this part of the algorithm is not necessary. As noted by Kitaev (*12*), if only the classical information of the QFT (such as the period r) is of interest, $2n$ qubits subject to a QFT can be replaced by a single qubit. Still, this approach requires qubit recycling (specifically, in-sequence single-qubit readout and state reinitialization) paired with feed-forward behavior to compensate for the reduced system size.

In the following, Kitaev’s QFT will be referred to as KQFT^(M). It replaces a QFT acting on M qubits with a semiclassical QFT acting repeatedly on a single qubit. Similar applications of Kitaev’s approach to a semiclassical QFT in quantum algorithms have been investigated (*13–15*). For the implementation of Shor’s algorithm, Kitaev’s approach provides a reduction from the previous n computational qubits and $2n$ QFT qubits (in total, $3n$ qubits) to only n computational qubits and 1 KQFT⁽²ⁿ⁾ qubit (in total, $n + 1$ qubits).

The second key ingredient of Shor’s algorithm—and a notably more challenging aspect than the QFT—is modular exponentiation, which admits the following general simplifications.

1) Considering Kitaev’s approach (Fig. 1), the input state $|1\rangle$ (in decimal representation) is subject to a conditional multiplication based on the most significant bit k of the period register. At most, there will be two results after this first step.

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